

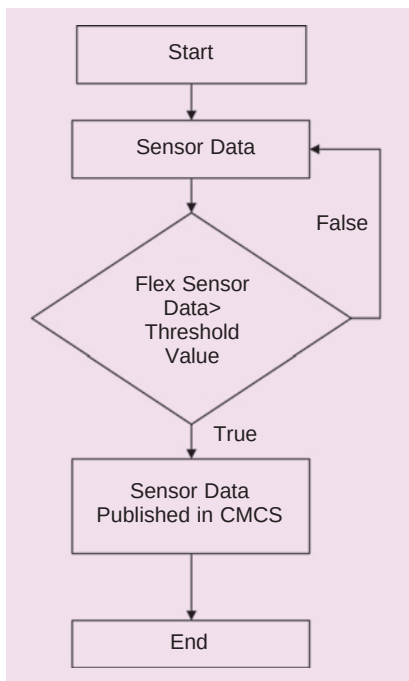


Fig. 2: Flow chart of the device working

opportunities to mitigate risks and enhance public safety. A proactive approach can prevent accidents, reduce financial losses, and create safer urban environments.

Detecting dangling wires using flex sensors

The presence of dangling wires in public spaces is a major safety concern. To address this issue, a methodology is proposed for detecting dangling wires using flex sensors and transmitting real-time data to a centralised monitoring system. This approach aims to improve public safety by identifying and addressing hazardous wires promptly.

Flex sensors are ideal for this application due to their thin and flexible nature, making them easy to attach to poles in public spaces. These sensors can detect changes in angle or position and can accurately determine the degree of tension on the hanging portion of the wire. By utilising flex sensors to detect the angle of wires in poles, a centralised monitoring system can be developed

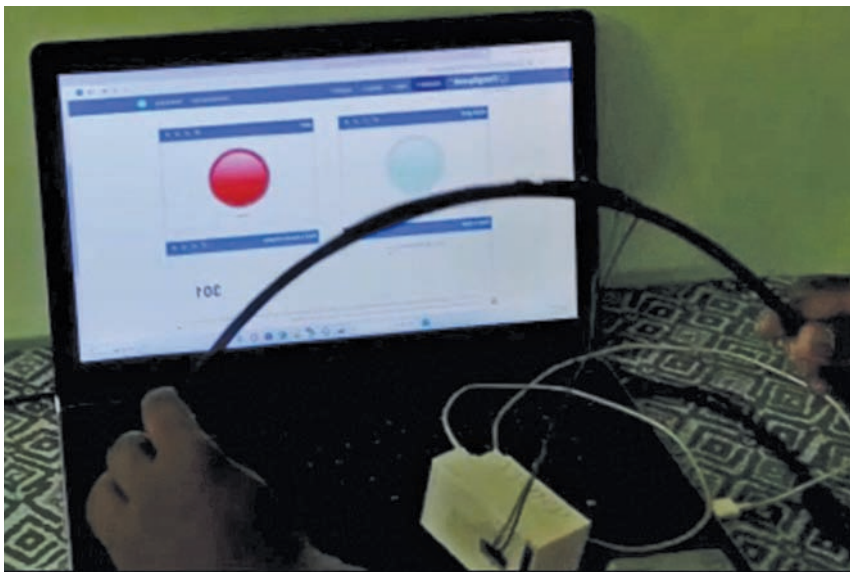


Fig. 3: GUI of the centralised monitoring station

to identify critical situations where the wires are in danger of snapping or causing harm. This system will help to prevent accidents and reduce the risk of injury to the public. The monitoring system can be designed to provide real-time information to the corporation or the head office of public affairs using mail and voice-over protocols.

The proposed monitoring system is designed to provide real-time updates to relevant authorities via email and voice-over protocols, enabling swift intervention. Compared to traditional wire detection methods, this system offers several advantages:

Cost-effectiveness. Flex sensors are inexpensive and easy to install.

Real-time monitoring. The system continuously tracks wire status, ensuring proactive intervention.

Automated alerts. Immediate notifications are sent to authorities when a wire is in a critical state.

Centralised monitoring and response system

The proposed system, as illustrated in Fig. 2, involves establishing a

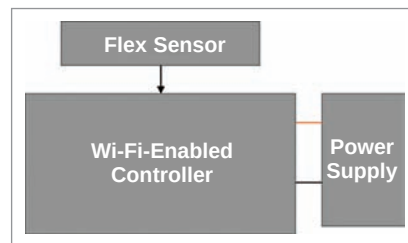


Fig. 4: Block diagram of the dangling wire detection system

centralised monitoring hub within the corporation or public affairs office. This hub receives real-time data from flex sensors, enabling authorities to assess and respond to critical wire conditions. The system can be designed using mail and voice-over protocol to inform the concerned authorities about the status of the wire. This will enable the authorities to immediately address the issue and prevent accidents caused by dangling wires. The system can be configured to send alerts based on severity:

Immediate alerts. If a wire is in a hazardous state, responsible local authorities receive urgent notifications.

Regular reports. Higher authorities receive scheduled reports to track ongoing maintenance and